MODULE 30 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

Cost Visibility, Chargeback and Alerting

Get detailed billing data where you can query it, charge shared platforms back fairly, including AI tokens, and hear about overspend before the invoice.

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Showback or chargeback?

SHOWBACK

Show teams what they used

- Reports, no money moves
- Builds awareness
- Tolerates rough allocation
- A common first step

CHARGEBACK

Bill teams for what they used

- Costs move between budgets
- Needs defensible allocation
- Shared costs must be split fairly
- Drives real behaviour change

THE TIEBREAKER

Chargeback needs allocation that teams will accept: measured usage, and fair rules for shared costs.

Where cost data lives

TOOL	GRANULARITY	BEST FOR
Cost Explorer	Daily or monthly, some hourly	Interactive analysis, forecasts
Data Exports: CUR 2.0	Hourly line items, resource IDs	Detailed analysis in Athena or BI
Data Exports: FOCUS	Open billing standard schema	Multi-cloud FinOps tools
Cost categories	Any of the above	Business-unit views and rules

ON THE EXAM

Line-item, resource-level detail for your own queries: CUR 2.0 through Data Exports.

Split cost allocation data

- **ECS tasks and EKS pods** get their own cost lines in CUR.
- **Based on** each container's share of the instance's CPU and memory.
- **EKS accelerated instances:** GPU, Trainium and Inferentia shares too.
- **Kubernetes attributes as tags:** cluster, namespace, workload.
- **Enable** in the management account's cost management preferences.

> ON THE EXAM
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> One shared cluster, many teams, billed by what their pods consumed: split cost allocation data.

Monitoring Bedrock token usage

NEED	MECHANISM
Tokens by model, live	CloudWatch metrics: input and output token counts
Cost by application	Application inference profiles with cost allocation tags
Cost by caller	IAM principal cost allocation, with tagged roles
Per-request detail	Model invocation logs with token counts

ON THE EXAM

Several applications on one model, charged back separately: a tagged inference profile per application.

Cost categories and split charges

- **Rules** map accounts, tags, services and other categories to values.
- **Split charge rules** share a cost across values: even, proportional, or fixed.
- **Typical shared costs**: support, platform accounts, untagged spend.
- **Visible** in Cost Explorer, Budgets and CUR.

> ON THE EXAM
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> Distribute shared costs like support fees across business units: split charge rules.

Budgets or Cost Anomaly Detection?

AWS BUDGETS

Did we cross a line?

- Thresholds you set: cost, usage, coverage
- Actual or forecasted alerts
- Budget actions: IAM or SCP policy, stop instances
- Good for known limits

COST ANOMALY DETECTION

Is this unusual?

- Machine learning on spend patterns
- Managed monitors by service, account, tag, category
- Root cause, with AI investigations
- Good for unknown surprises

THE TIEBREAKER

A known limit: Budgets. A surprise nobody set a threshold for: Cost Anomaly Detection. Use both.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Hourly, resource-level data in Athena	CUR 2.0 through Data Exports
Teams share one Kubernetes cluster	Split cost allocation data
Apps share one Bedrock model	Tagged application inference profiles
Forecast exceeds a limit, act automatically	Budgets with budget actions
Unexpected spikes, every team, no thresholds	Cost Anomaly Detection
Spread support fees across units	Cost category split charges

ON THE EXAM

Detail is CUR, sharing is split allocation, limits are Budgets, surprises are anomalies.

> Meridian Freight's FinOps team wants hourly cost data at resource level for all accounts, queried with SQL and joined to the company's own project database.
>
> The data must be refreshed automatically.

What should the solutions architect configure?

A Cost Explorer saved reports

B A CUR 2.0 export through Data Exports to S3, queried with Athena

C Monthly billing PDFs

D AWS Budgets reports

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Cost Explorer

Interactive and aggregated; not raw line items for SQL joins.

B CUR 2.0 to S3

Hourly line items with resource IDs, refreshed, queryable in Athena.

C Billing PDFs

Summary documents, not data.

D Budgets reports

Track budgets, not line items.

THE SIGNAL

SQL over line-item, resource-level, hourly data: CUR 2.0.

> Eight Meridian Freight teams share one EKS cluster, each in its own namespace, including GPU nodes used by the data science team.
>
> Finance wants to charge each team for the compute its pods actually reserved, including GPUs, without separate clusters.

What should the solutions architect enable?

A Split cost allocation data for EKS, charging back by the namespace tag

B A cost allocation tag on each node

C Divide the cluster cost evenly by eight

D A separate account per team

◆ PAUSE AND DECIDE

What should the solutions architect enable?

A Split cost allocation

Pod-level costs from CPU, memory and GPU reservations, tagged by namespace.

B Node tags

Nodes run pods from many teams.

C Even split

Not what each team actually used.

D Account per team

Means separate clusters, which is ruled out.

THE SIGNAL

Shared cluster, charge by what pods used, GPUs included: split cost allocation data.

> Four Meridian Freight applications in one account call the same Bedrock model. Finance must charge each application's owning team for its token spend.
>
> The team wants the split to appear in Cost Explorer without parsing logs.

What should the solutions architect implement?

A Move each application to its own AWS account

B Use CloudWatch token metrics grouped by model ID

C Parse model invocation logs nightly

D Create an application inference profile per application, with an activated cost allocation tag

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Account per application

Works, but a large restructure for a billing view.

B Metrics by model

All four apps share the model, so they share the metric.

C Parse logs

Custom work the team wants to avoid.

D Tagged inference profiles

Each application's calls carry its own tag into billing.

THE SIGNAL

Many applications, one model, split in billing: tagged application inference profiles.

> Meridian Freight's development accounts have a monthly limit of 20,000 dollars. The team wants a warning when spend is forecast to exceed the limit.
>
> If actual spend reaches the limit, new resources must be blocked automatically until next month.

What should the solutions architect configure?

A Cost Anomaly Detection with an email subscription

B A CloudWatch billing alarm

C An AWS Budget with a forecasted alert, and a budget action that applies a restrictive SCP at 100% of actual spend

D A weekly Cost Explorer report

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Anomaly Detection

Flags unusual spend, not a fixed limit.

B Billing alarm

Alerts on actual charges only; cannot apply an SCP.

C Budget with action

Forecast warning, and an automatic SCP when the limit is hit.

D Weekly report

Neither timely nor automatic.

THE SIGNAL

A known limit, a forecast warning and an automatic response: Budgets with budget actions.

> Meridian Freight has 40 teams, identified by a Team cost allocation tag, and new teams appear every quarter. Last month one team's mistake doubled its spend for a week before anyone noticed.
>
> The company wants unusual spend for any team flagged quickly, with its likely cause, without maintaining thresholds.

What should the solutions architect configure?

A. Cost Anomaly Detection with an AWS managed monitor on the Team tag, alert subscriptions, and investigations with Amazon Q

B. One AWS Budget per team with fixed thresholds

C. A CloudWatch alarm per team

D. Monthly Cost Explorer reviews

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Managed anomaly monitor

Learns each team's pattern; new tag values monitored automatically.

B Budget per team

Forty thresholds to maintain, and more every quarter.

C Alarm per team

Thresholds again, built by hand.

D Monthly reviews

Weeks too slow.

THE SIGNAL

Unusual, any team, no thresholds, new teams included: Cost Anomaly Detection managed monitor.

> Meridian Freight bills four business units monthly. Enterprise Support fees and a shared networking account cannot be tagged to any one unit.
>
> Finance wants these shared costs distributed in proportion to each unit's own spend, shown in Cost Explorer.

What should the solutions architect configure?

A. Tag the support charges with each business unit

B. A business-unit cost category with split charge rules distributing the shared costs proportionally

C. A separate organisation per business unit

D. Leave shared costs unallocated

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Tag support charges

Support fees are not resources and cannot carry tags.

B Split charge rules

Shared costs distributed proportionally, visible in Cost Explorer.

C Separate organisations

Duplicates support and networking.

D Unallocated

Chargeback then misses real costs.

THE SIGNAL

Costs that belong to everyone, split by rule: cost category split charges.

What to carry out of this module

- **CUR 2.0** for detail; **Cost Explorer** for interactive analysis.
- **Split cost allocation** for shared clusters, GPUs included.
- **Bedrock chargeback** by tagged inference profiles or IAM principals.
- **Budgets** for known limits and actions; **anomaly detection** for surprises.
- **Split charges** distribute costs nobody owns.

> ON THE EXAM
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> Measure shared usage, split shared costs by rule, and alert on both limits and surprises.